KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

Astrometric analysis — summary

Prepared 2026-09-06. Solution: `~/KMTdata/Results/v3_Final/IFfinal_260058_CTIO_<field>.mat`

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1. Data

quantity	BLG41	BLG01
exposures	19 133	19 981
after cuts	17 354	17 687
sources fitted	594	621
nights	1 985	2 009

KMT CTIO, 2016–2026, I band. Two **overlapping cut-outs** of the same star, 300 × 300 pix at 0.4″/pix (2′ × 2′). They share **no exposures**: the two fields are observed alternately, a median of 12.2 min apart, on 1 947 common nights, so they are independent measurements.

V-band frames exist for 2016–2022 (~1 340 per field) and are used only to check the colour; they are not used in the astrometry.

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2. What is in the fit

Per epoch

- full affine transformation, 6 parameters: translation, rotation, scale, shear

Per source

- position (x_0, y_0) and proper motion (μx, μy) — 4 parameters

Detrending terms

- **differential chromatic refraction (DCR)** — `[1, sin(pa)·secz, cos(pa)·secz]` plus higher orders in `secz·sin(pa)` and `secz·cos(pa)`, fitted **in 6 colour bins** (equal population).
- **annual term**
- **pixel-phase correction**, from the per-epoch registration shifts
- **SysRem**, 2 components on the astrometric residuals, computed **once over the whole decade** and reused by every sub-fit
- iterative reweighting from the residuals, with moving-median outlier rejection

Convergence. 15 weighted iterations before SysRem, 6 after. The last three iterations move the bright-star median by 0.003 mas (BLG41) and 0.001 (BLG01).

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3. What is not in the fit

- **parallax** — bulge sources give ≈0.12 mas, far below the noise floor
- **any microlensing or astrometric-deviation model** — the solution is model-free; no theoretical curve of any kind has been fitted or subtracted
- **absolute astrometry inside the fit** — the solution itself is *relative*: proper motions are in the registered pixel frame, 400 mas/pix is assumed rather than measured, and the orientation is not used. A Gaia tie is available afterwards (`ml.util.gaiaTie`, see section 7) and gives the orientation, the scale and absolute proper motions, but it is applied to the finished solution and does not constrain the fit
- acceleration or binary motion
- any correction for the faint-end photometric bias (§6)

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4. Steps

1. **`KMT_pipelineI`** — registration, PSF photometry, source matching → `MatchedSources`; cross-match to the OGLE catalogue for I and V, using a **per-field** pixel offset (BLG41 229.202/229.283, BLG01 227.542/229.689). 2. **`ml.util.mmsFromMatchedSources`** — magnitudes rebuilt from `FLUX_PSF` (the pipeline leaves every `MAG_*` column empty), per-epoch zero point anchored on OGLE I, then SysRem photometry; airmass and parallactic angle computed from JD and the field centre; colour from OGLE V−I; quality cuts. 3. **Joint fit 2016–2025** → proper motions 4. **Full-decade fit** with those motions held → one SysRem correction for the run. 5. **Ten single-season fits** reusing that correction. 6. **Final global fit, 2016–2026, motions solved.**

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5. Calibrating stars — selection, and their colour in the plots

Selection (all conditions required):

- **14 < I_OGLE < 19**
- **no companion brighter than I = 18 within 2.0 arcsec**, i.e. **5 KMT pixels** at 0.4″/pix. The companion list is the OGLE catalogue, which is at 0.26″/pix and resolves about six times more sources than KMT detects, mapped into KMT pixels with the per-field offset above.
- a star's own OGLE counterpart is excluded from its companion test.

Surviving: **287 stars in BLG41, 253 in BLG01.**

Colour key in the RMS plots

colour	meaning
grey dots	all field stars
blue circles	**calibrating stars**, as selected above
black line	running median of *all* stars, in 0.5 mag bins
red star	the target

One point to be clear about: in this solution the per-epoch transformation is fitted from **all** sources, each weighted by its own residual scatter. The calibrating stars are a clean, well-measured subset shown for reference; they do not exclusively define the frame. `UseRefSources`, which would restrict the frame fit to them, is available but is **off** here.

Why the frame is fitted from all stars. This was measured rather than assumed. `UseRefSources` was run at full scale on both fields, selecting on a magnitude window and on having no companion brighter than I = 18 within a given radius, everything else at the defaults (runs in `~/KMTdata/Results/AstrometryMSc_ref/`):

run	ref stars	rstd bright X/Y	seasonal wander
BLG41, frame from all sources	594	**8.85** / **9.23**	**2.21** / **2.57**
BLG41, companion radius 5 pix	36	9.06 / 9.73	1.97 / 2.80
BLG41, companion radius 3 pix	48	8.67 / 9.91	1.96 / 2.82
BLG01, frame from all sources	621	**6.85** / **7.40**	~1.5 / 1.3
BLG01, companion radius 5 pix	10	**17.27** / **14.55**	5.74 / 3.55

mas. **BLG01 fails outright** — two and a half times worse, with the field's seasonal wander nearly quadrupled. **BLG41 is roughly neutral**, not clearly worse: the bright rstd degrades slightly while the event source's seasonal wander improves by 11% in X. So the honest summary is that restricting the frame never helped and once hurt badly, not that it is uniformly worse.

The reference stars themselves are excellent, which is what makes the result informative. On BLG41, 36 survivors are measured to **4.2** / **4.0 mas and detected in 100% of epochs**, against **14.0** / **14.8 mas and 81%** for the field as a whole. The selection did exactly what it was meant to do.

But the frame is an average, and its error falls as the star scatter over the square root of their number:

	scatter per star	N	frame noise, σ/√N
BLG41, all sources	14.0 mas	594	**0.57 mas**
BLG41, 36 references	4.2 mas	36	**0.70 mas**
BLG01, 10 references	~4.2 mas	10	**1.33 mas**

The clean stars are 3.3 times better individually, but there are 4.1 times fewer of them in the square root, so the crowd wins — narrowly on BLG41, which is why that field came out neutral, and decisively on BLG01. The last row predicts a degradation of 1.33 / 0.56 = **2.4x**, against the **2.5x** actually measured. Ten reference stars are twenty equations for six parameters per epoch, and their own 4 mas scatter propagates into the frame and from there into every source. (The BLG01 row assumes BLG41's measured reference-star precision for its ten stars, which was not measured separately.)

Three further reasons the subset cannot win on these data:

- **The fit already does this, more gently.** Each source is weighted by its own residual scatter — `calculateNee` takes `median(W,1,'omitnan')` per source — so a noisy faint star already contributes almost nothing. `RefSrcFlag` replaces that soft weighting with a hard cut, discarding information the weights were using correctly.
- **The dominant noise is shared.** The two-field comparison put about 74% of the nightly noise in the atmosphere, common to every star. Reference stars sit under the same atmosphere as the target, so a cleaner frame cannot reach it. Only the ~26% reduction noise is attackable, and 26% in variance is 15% in amplitude — which caps the possible gain at about the size of the shifts seen above. Nothing here could have been a factor of two.
- **The selection is not a neutral subset.** It requires OGLE coverage, and OGLE covers only 29% of BLG01's sources, leaving 13 candidates there before the isolation cut. BLG01's failure follows from colour coverage rather than from the idea.

The approach is sound and remains implemented; on these data the reference frame was simply not the binding constraint — the atmosphere was.

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6. The event is not confined to one season

With the OGLE parameters t_0 = JD 2461214.2 (2026-06-22), t_E = 231.7 d and u_0 = 0.499, the magnification at each season's mean epoch is as below.

- **u** is the angular separation between lens and source at that epoch, in units of the Einstein radius: `u = sqrt(u₀² + ((t − t₀)/t_E)²)`. It is dimensionless, falls to u_0 = 0.499 at closest approach, and is large long before and after the event.
- **A** is the **flux** magnification, `A = (u² + 2) / (u·sqrt(u² + 4))`. A = 1 means unmagnified; A = 2 means twice as much light.
- **Δmag** is the same quantity in magnitudes, `Δmag = −2.5·log₁₀(A)`. The two columns are not independent: A = 2.067 is Δmag = −0.788, and −0.788 mag inverts to A = 2.066. Where the text below quotes "a magnification of about 2.1" and "0.8 mag" it is quoting the same number twice.

season	($t - t_0$) [d]	u	A	Δmag
2016	−3644	15.74	1.000	0.000
2019	−2564	11.08	1.000	0.000
2022	−1474	6.38	1.001	−0.001
2023	−1114	4.83	1.003	−0.003
2024	−754	3.29	1.012	−0.013
2025	−394	1.77	**1.085**	−**0.089**
2026	−44	0.53	**2.067**	−**0.788**

The event is above 1% magnification for **4.4 years**, above 5% for 2.6 years and above 10% for 2.0 years.

The model predicts 0.80 mag of brightening at the 2026 season median, whereas the measured magnitudes give 1.085 (18.724 at baseline against 17.639 in 2026). The 0.29 mag difference is the faint-end photometric bias of section 7, which is +0.39 mag at I ≈ 18.1 and +0.13 at I ≈ 17.3: a *difference* of two measured magnitudes inflates by the difference of their biases, 0.80 + 0.26 = 1.06 against 1.085 observed. It is a consistency check on the bias, not a discrepancy with the model.

Calling 2026 "the event season" and 2016–2025 "pre-event" is therefore wrong: **2025 is already magnified by 9%** and 2024 by 1.3%.

This matters for the proper-motion baseline. The astrometric deviation of a microlensing event does not peak with the photometric one: for these parameters it is largest near $u = \sqrt{2}$, which falls in **2025** (2.46 mas), with 2024 at 1.91 mas and 2026 back down to 1.63 mas as the source passes closest approach. So the joint fit that supplies the proper motions, which holds out 2026 alone, still contains the two seasons carrying the **largest** astrometric signal.

Any proper motion quoted here is therefore fitted through the event, and would absorb part of a real astrometric excursion. A baseline free of it would have to stop around 2022, which costs a third of the time span and most of the leverage on the proper motion. Nothing in this report depends on that choice — no astrometric model is fitted — but a future attempt to measure the deviation must not treat 2016–2025 as an event-free baseline.

7. Caveats that matter for reading the plots

The measured magnitudes run faint at the faint end. `KMT − I_OGLE` is flat to ±0.04 down to I = 17, then rises to **+0.50 by I = 18.75**. The target's own measured magnitude is 18.70 against its catalogue value of **18.13**. All plots here therefore use the **OGLE catalogue I** on the magnitude axis. PSF and aperture photometry bracket OGLE and disagree by ~0.96 mag at I = 18.5, which points to crowding: OGLE lists 3 783 sources in this cut-out where KMT detects 594. **The astrometry is not affected** — positions come from the PSF centroid, not the flux, and removing the photometric anchor entirely was measured to cost 0.012 mas.

The CSV carries two error options; choose deliberately.

column	what it is
`errX_decade`, `errY_decade`	the black running-median curve of the RMS plot read at the target's catalogue magnitude I = 1 **Constant.**
`errX_season`, `errY_season`	the target's **own** residual RMS within its observing season. Follows its brightness.

Neither is a per-epoch measurement uncertainty. The decade value is a population estimate at a fixed magnitude; the season value is the target's own scatter.

They differ by a factor of two where it matters most. The target is at I ≈ 18.1 for nine seasons and is magnified by 0.80 mag in 2026, so:

season	target mag	errX / errY, BLG41	BLG01
1 (2016)	18.73	19.28 / 18.22	19.32 / 19.51
4 (2019)	18.71	31.92 / 26.31	29.21 / 26.56
6 (2022)	18.74	33.81 / 27.17	38.55 / 31.45
9 (2025)	18.61	16.85 / 15.13	16.76 / 14.75
10 (2026)	**17.65**	**12.46 / 11.38**	**11.82 / 11.54**
decade constant	—	21.80 / 23.48	22.68 / 22.63

Using the constant would overstate the uncertainty at the peak of the event by about a factor two, and understate it in 2022 by a third.

The sign of a proper-motion component carries no physical meaning. The fitted motion of the target is +2.20 / −2.31 mas/yr in BLG41 and −0.66 / −2.90 in BLG01: the same star, the same nights, opposite signs in X. This is a gauge difference between the two frames, not anything about the source. The per-epoch transformation is free at every epoch, so it absorbs any motion common to the field, and what survives is defined only relative to the mean motion of each cut-out's own star population. Those populations differ:

quantity	BLG41	BLG01
median PM of the field's own sources	+2.128 / +0.183	−0.261 / −0.617

Measured directly on the 383 well-measured stars common to both fields, the disagreement is the same for every star:

quantity	X	Y
PM difference, BLG01 − BLG41, all common stars	−2.41 ± 0.43	−0.60 ± 0.57

the target	−2.86	−0.59

The target's offset is the population's offset. Fitting the difference as a constant plus a linear term in position gives, in addition, a relative rotation of 0.28 deg/yr and a scale drift of 4.7e-3 /yr between the two frames; removing both leaves 1.14 / 0.35 mas/yr, consistent with per-star measurement error.

So the Y values (−2.31 and −2.90) agree only because the Y gauge offset is small, and the X values disagree only because the X offset is large. **What is physical is the differential motion between stars within one field**, unless the frame is tied to an external one — which it now can be.

Tied to Gaia, the two fields agree. `ml.util.gaiaTie` carries a solution onto the Gaia frame through OGLE, giving the orientation, the plate scale and an absolute proper motion for every matched source:

quantity	BLG41	BLG01
KMT sources matched to Gaia	532 of 594 (90%)	447 of 621 (72%)
median separation	0.259 arcsec	0.358 arcsec
calibrators (Gaia PM error < 0.4 mas/yr, I < 18)	426	327
fitted rotation of the pixel axes on the sky	**+131.01 deg**	**+130.79 deg**
gauge offset removed	−0.36 / +4.41 mas/yr	−2.84 / +3.88
scatter about the Gaia relation	3.75 / 3.83 mas/yr	3.38 / 3.51

The target's proper motion:

frame	BLG41	BLG01	difference
relative (detector x, y), as fitted	+2.196 / −2.310	−0.660 / −2.900	2.856 / 0.590
absolute (μ_α cos δ, μ_δ), Gaia frame	**−2.917 / −7.248**	**−2.215 / −7.244**	**0.702 / 0.004**

mas/yr, taken from `IF.ParS(3:4,Ie)` at 400 mas/pix, target index 383 (BLG41) and 450 (BLG01). **The two rows are not in the same axes**: the relative values are in the cut-out's detector x/y, the absolute ones are (μ_α cos δ, μ_δ), since `gaiaTie` returns `Tie.A \ (PM - Offset)` and so carries the pixel frame into the Gaia frame through a 131 deg rotation with parity and a 0.92 scale.

The sign disagreement disappears once both are on the same physical frame, and the two rotations, derived independently from separately reduced data, agree to 0.22 deg — which is what makes the tie credible rather than a fit to noise. The relative disagreement itself is gauge, not astrophysics: both cut-outs have nearly the same orientation (131.01 and 130.79 deg), so the +2.20 against −0.66 in x is the unconstrained constant-plus-gradient freedom of each field's own frame, which is exactly what the tie removes.

Against Gaia's own value for this star. The target has a Gaia counterpart, so the absolute motion can be checked directly rather than only field against field:

	μ_α cos δ	μ_δ
Gaia DR3, the target	−2.479	−9.129
ours, BLG41	−2.917 (−0.44)	−7.248 (+1.88)
ours, BLG01	−2.215 (+0.26)	−7.244 (+1.89)

mas/yr, with the difference from Gaia in brackets. μ_α cos δ agrees well. μ_δ sits 1.88 mas/yr away, and by nearly the same amount in both fields — which should **not** be read as two independent confirmations, because BLG01 reused BLG41's OGLE-to-Gaia solution and a common systematic in that bridge would reproduce this pattern exactly. Against the per-source scatter it is about 0.5 sigma and unremarkable. It is not the event: 1.88 mas/yr over the decade baseline is some 19 mas of accumulated displacement, an order of magnitude beyond the ~2.5 mas peak deviation the model allows.

Which error applies. Three different numbers circulate and they measure different things:

- **3.4 to 3.8 mas/yr** — scatter about the Gaia relation, *per source*. This is the one that applies to a single star's absolute proper motion, the target included.
- **0.16** / **0.17 mas/yr** — the error on the tie *transformation*, from 532 and 447 matched sources. It bounds the frame, not any individual star.
- ~**1.0** / **0.7 mas/yr** — the target's own relative proper motion uncertainty, from the scatter of its season means about its fitted line.

Reduction choices contribute far less: the pixel-phase on/off variants give +2.21 / −2.31 and +2.27 / −2.27 (BLG41), −0.65 / −2.89 and −0.71 / −2.94 (BLG01), a spread of about 0.06 mas/yr.

Two cautions. The μ_δ agreement of 0.004 mas/yr between the fields is far better than either field's own precision and is therefore luck; the honest statement is that the two agree to within errors of order 1 mas/yr on the mean. And the tie fixes the **frame, not the precision**: our per-source scatter of 3.4 to 3.8 mas/yr exceeds Gaia's own spread of 2.9 to 3.3, so most of the per-star difference is ours and nothing in the residual analysis improves.

Plotted positions are relative to the target's own mean position, not to the field centre and not to any absolute reference. Each motion curve has the target's decade-average position subtracted, so the zero line is that average. The frame itself is set by the ensemble of 594 (BLG41) and 621 (BLG01) field stars through the per-epoch transformation, not by the geometric centre of the cut-out: differences along a curve are meaningful, the absolute level is not, and the level cannot be compared between the two fields, whose cut-outs have different origins and independently fitted frames. For scale, the target lies +0.42, +0.39 pix (+168, +155 mas) from the centre of the BLG41 cut-out and +1.55, +0.15 pix (+620, +60 mas) from the centre of BLG01; the centring absorbs both.

Only sources with an OGLE counterpart appear in the RMS plots (540 of 594 in BLG41, 478 of 621 in BLG01), which biases the plotted sample slightly bright.

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8. Where the numbers stand

quantity	BLG41	BLG01
bright-star residual (I < 17), ΔX/ΔY	6.414 / 6.879 mas	6.475 / 7.007 mas
calibrating stars, ΔX/ΔY	8.29 / 8.30	7.47 / 8.14
target, whole decade	24.62 / 21.00	25.48 / 21.78
target, 2026 season only	≈ 11.7 / 11.1	≈ 12.2 / 11.7

The target's decade figure is dominated by the seasons in which it sits near its baseline I ≈ 18.1; in 2026 the magnification of about 2.1 brightens it by 0.8 mag and its residual halves. 2025 is brighter than baseline too, by 0.09 mag. **12 mas is the target's 2026 precision, 24 mas its decade-average precision** — they are different quantities.

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9. Files in this report

file	contents
`report_RMS_afterPM.png`	residual RMS against OGLE I, per axis, per field — **after position and proper motion rem**
`report_motion_<field>_nobin.png`	source motion, unbinned; 2x2 panel, columns X and Y, top row = position with PM **re** residual
`report_motion_<field>_binsid.png`	the same, binned in **one sidereal month** (27.321661 d); error bars are the **RMS within the**
`source_motion_<field>.csv`	`JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, seaso`

Eight motion figures in total: 2 fields × 2 axes × 2 binnings.

Figures

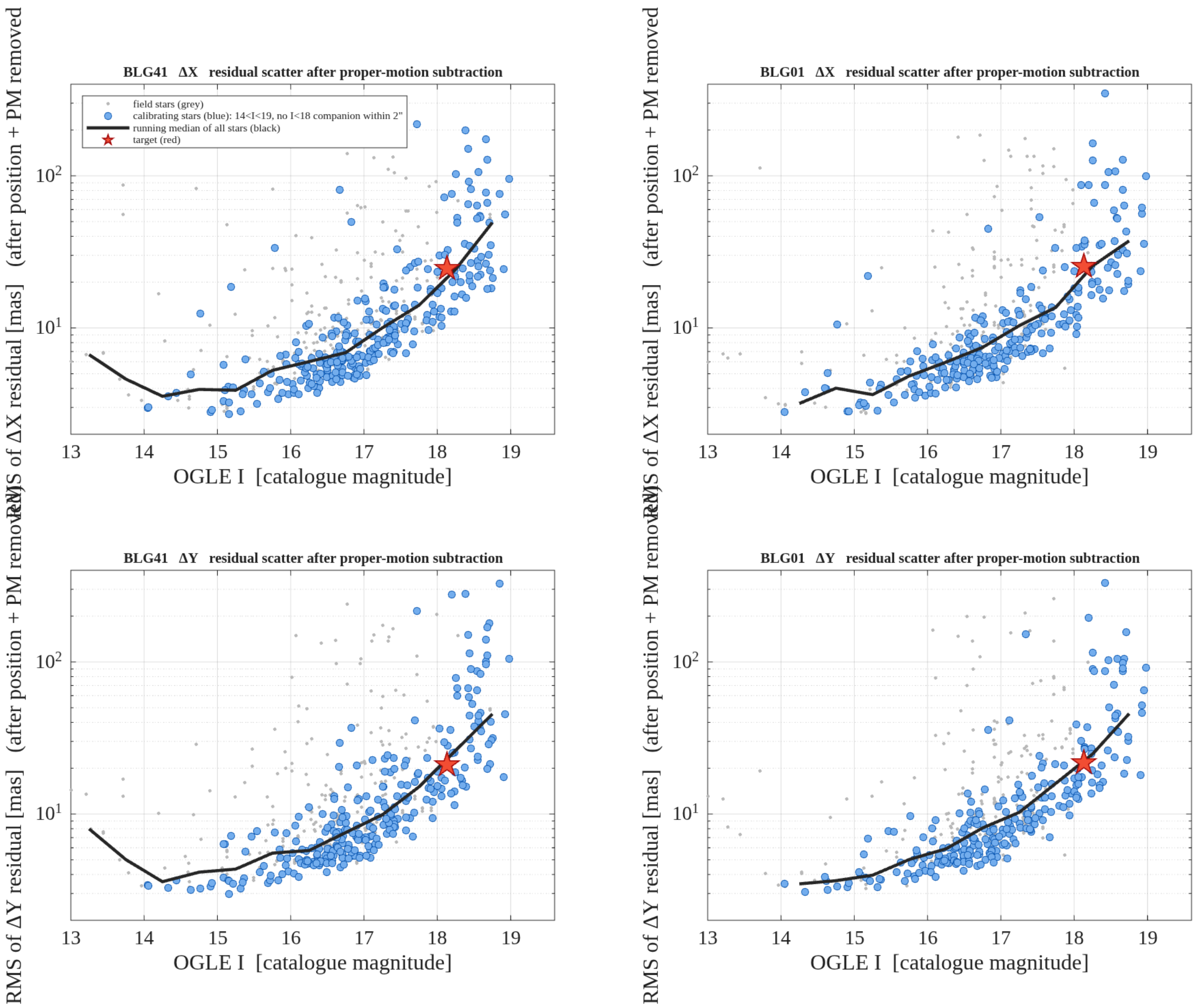

Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **after position and proper motion have been removed**. Grey: all field stars. Blue: calibrating stars (14<I<19, no I<18 companion within 2 arcsec). Black: running median of all stars, the curve the CSV decade errors are read from. Red: the target.
`report_RMS_afterPM.png`

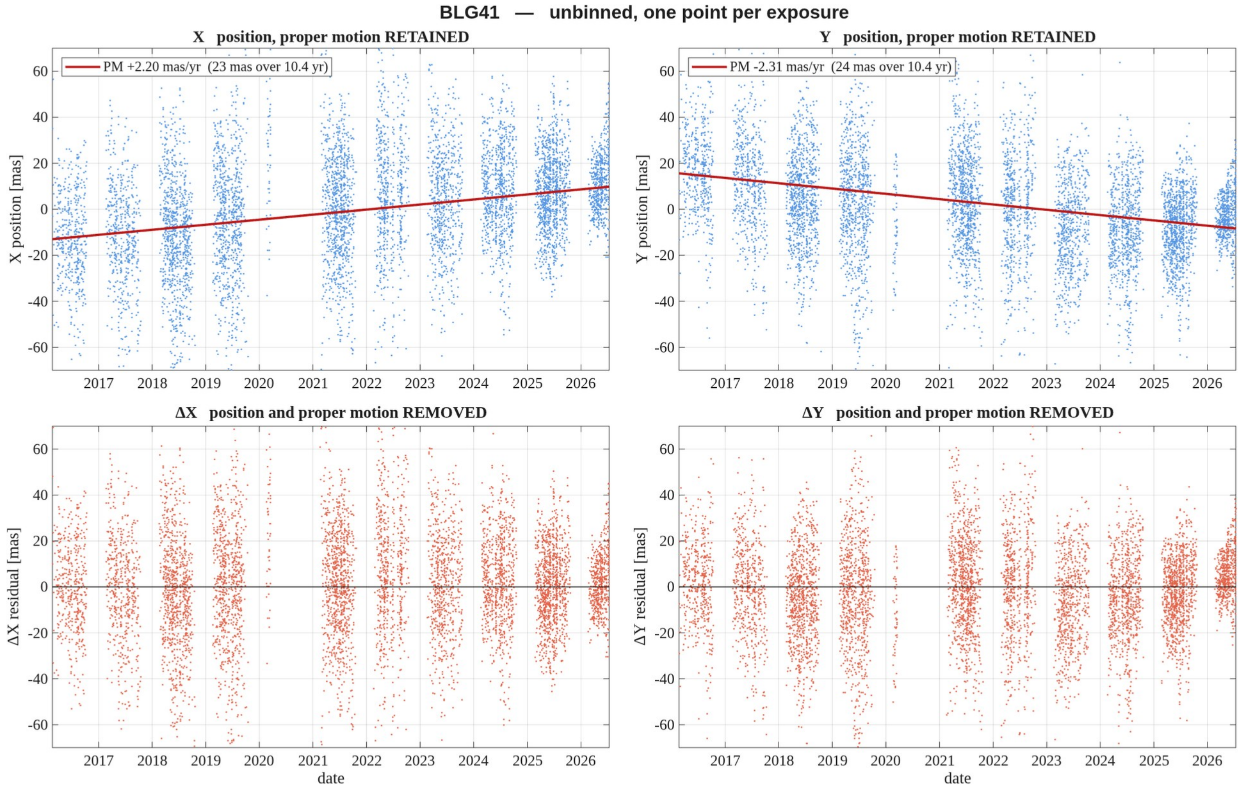

Figure 2. **BLG41** — unbinned, one point per exposure, no error bars. Columns are the two axes, **X left and Y right**. Top row: the measured position with the proper motion **retained**; the red line is the fitted proper motion. Bottom row: the same with position and proper motion removed. All four panels share one vertical scale so the axes can be compared directly; it is set from the robust scatter, so a handful of extreme points fall outside it. Positions are relative to the target's own mean.

`report_motion_BLG41_nobin.png`

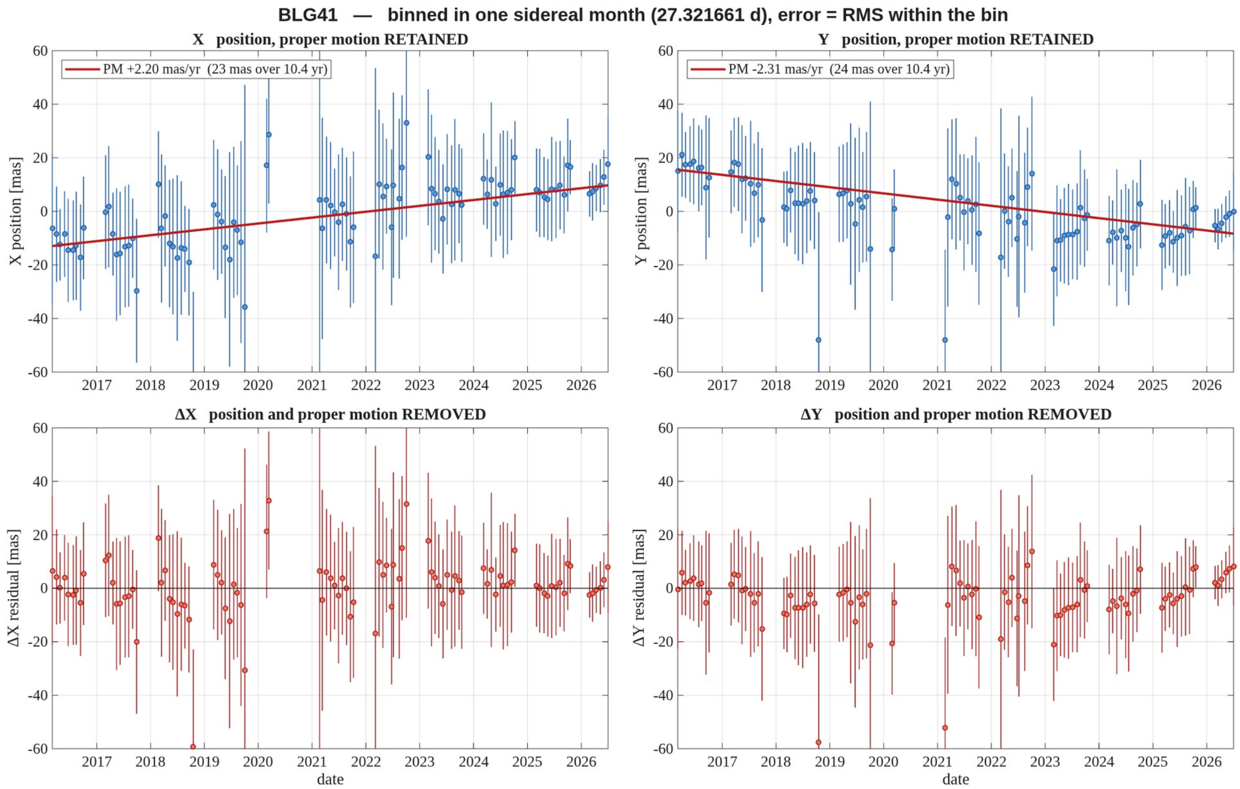

Figure 3. **BLG41** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**. Top row: the measured position with the proper motion **retained**; the red line is the fitted proper motion. Bottom row: the same with position and proper motion removed. All four panels share one vertical scale so the axes can be compared directly; it is set from the robust scatter, so a handful of extreme points fall outside it. Positions are relative to the target's own mean.

`report_motion_BLG41_binsid.png`

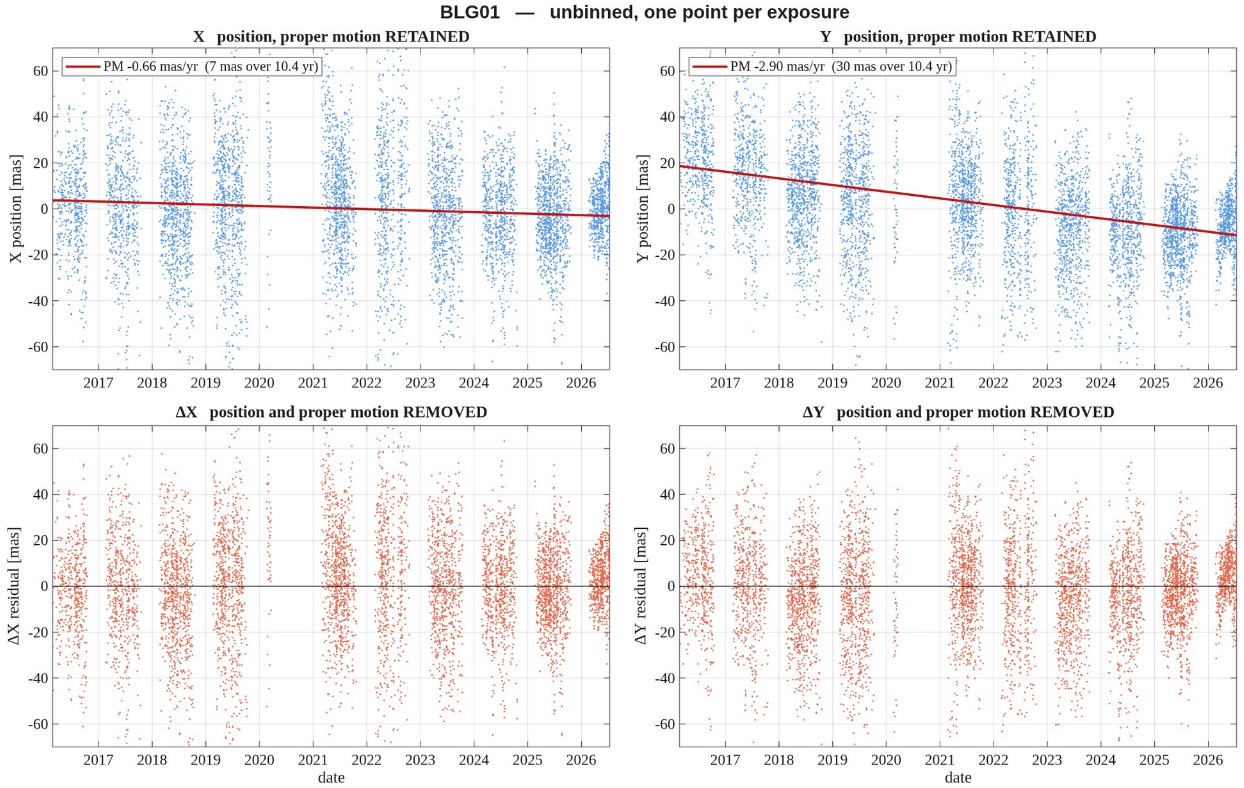

Figure 4. **BLG01** — unbinned, one point per exposure, no error bars. Columns are the two axes, **X left and Y right**. Top row: the measured position with the proper motion **retained**; the red line is the fitted proper motion. Bottom row: the same with position and proper motion removed. All four panels share one vertical scale so the axes can be compared directly; it is set from the robust scatter, so a handful of extreme points fall outside it. Positions are relative to the target's own mean.

`report_motion_BLG01_nobin.png`

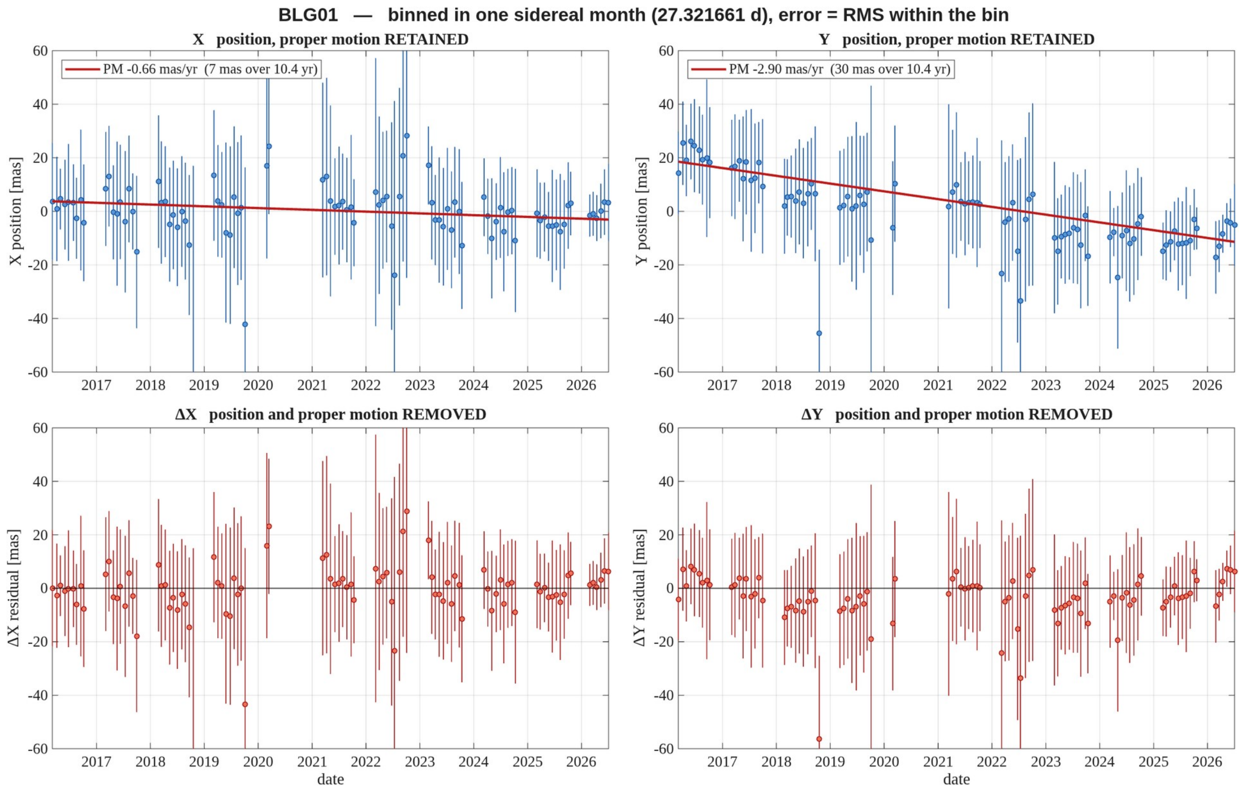

Figure 5. **BLG01** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**. Top row: the measured position with the proper motion **retained**; the red line is the fitted proper motion. Bottom row: the same with position and proper motion removed. All four panels share one vertical scale so the axes can be compared directly; it is set from the robust scatter, so a handful of extreme points fall outside it. Positions are relative to the target's own mean.

`report_motion_BLG01_binsid.png`